

Diagnosing Quadrant and Parameter-Orientation Errors

Find the mistake, fix the value, and say what went wrong in each conversion below.

Directions:

- Every problem below contains one student's work with exactly one thing wrong in it.
- Part (a) asks for the corrected value. Part (b) asks you to name the error in a sentence or two.
- Angles are in radians. Give exact values unless a decimal is asked for.

1. A student converts the rectangular point $(-3, 3)$ to polar form. For the angle they enter $\arctan\left(\frac{3}{-3}\right)$ on a calculator and write $\theta \approx -0.79$.

(a) Give the correct exact value of θ in $[0, 2\pi)$. $\theta =$ _____

(b) Name the error in the student's method.

- 2.** A second student converts $(-1, -\sqrt{3})$ to polar form the same way and reports $\theta \approx 1.05$.
- (a) Give the correct exact value of θ in $[0, 2\pi)$. $\theta = \underline{\hspace{2cm}}$
- (b) Name the error, and say why the two negative signs are what caused it.

- 3.** A student says the parametric curve $x = 2 \cos t$, $y = -2 \sin t$, $0 \leq t \leq 2\pi$, is the circle $x^2 + y^2 = 4$ traced counterclockwise from $(2, 0)$.

- (a) Find the point on the curve at $t = \frac{\pi}{2}$. $x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$
- (b) Use that point to name the error in the student's description.

4. A student writes: “for $x = t + 1$, $y = t^2$, the curve is traced from right to left as t increases.”

(a) Find x at $t = -2$ and at $t = 1$. $x = \underline{\hspace{2cm}}$ $x = \underline{\hspace{2cm}}$

(b) Name the error in the student’s claim.

5. For $x = \sqrt{t}$, $y = t - 3$, $t \geq 0$, a student eliminates the parameter, gets $y = x^2 - 3$, and draws the entire parabola.

(a) Find y when $x = 2$. $y = \underline{\hspace{2cm}}$

(b) Name the error: describe exactly which part of the drawn parabola is not on the parametric curve, and why.

6. A student converts the polar point $\left(-4, \frac{\pi}{3}\right)$ to rectangular form and writes $x = 2$, placing the point in the first quadrant.

(a) Find the correct coordinates. $x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$

(b) Name the error and say which quadrant the point really lies in.

7. A student claims that on the rose $r = 4 \cos(2\theta)$ the value of r is never negative, so “every petal points the way its angle says”.

(a) Find every θ in $[0, 2\pi)$ for which $r = 0$. $\theta = \underline{\hspace{2cm}}$

(b) Explain what happens to the graph for the angles between the first two of those values.

8. For $x = 2 + 3 \cos t$, $y = -1 + 3 \sin t$, a student eliminates the parameter and writes $(x - 2)^2 + (y + 1)^2 = 3$.

(a) Find the point on the curve at $t = 0$. $x = \underline{\hspace{2cm}}$ $y = \underline{\hspace{2cm}}$

(b) Show that this point does not satisfy the student's equation, and write the corrected equation.